
3AL Tutorial 6 2024 Solutions

1. Let G be a group acting on a set X and let $a, b \in G$ and $x, y \in X$. Prove the following:

- (a) If $a * x = a * y$, then $x = y$,
- (b) $a * x = b * y$ if and only if $(b^{-1}a) * x = y$.

(a) Suppose $a * x = a * y$. Then

$$a^{-1} * (a * x) = a^{-1} * (a * y)$$

$$1 * x = 1 * y$$

$$x = y.$$

(b) Suppose that $a * x = b * y$, then

$$b^{-1} * (a * x) = b^{-1} * (b * y)$$

$$(b^{-1}a) * x = 1 * y$$

$$(b^{-1}a) * x = y.$$

The converse is similar.

2. Prove that if H is a normal subgroup of G , then G acts on both H and G/H by conjugation.

NOTE: For $aH \in G/H$ and $g \in G$, we define $g(aH)g^{-1} = \{gahg^{-1} : h \in H\}$. To prove that conjugation is well-defined on G/H , you will have to show that $g(aH)g^{-1}$ is indeed an element of G/H .

Consider the map $* : G \times H \rightarrow H$ defined by $g * h = ghg^{-1}$. Note that because H is normal, we have $g * h \in H$ and so the map is well-defined. Moreover, we have $1 * h = 1h1^{-1} = h$. For $g_1, g_2 \in G$, we also have

$$g_1 * (g_2 * h) = g_1 * (g_2hg_2^{-1}) = g_1g_2hg_2^{-1}g_1^{-1} = (g_1g_2)h(g_1g_2)^{-1} = (g_1g_2) * h.$$

Therefore, conjugation is a group action by G on H .

Consider the map $*$: $G \times G/H \rightarrow G/H$ defined by $g * aH = g(aH)g^{-1}$. First, we must show that $g(aH)g^{-1} \in G/H$. Let $gahg^{-1} \in g(aH)g^{-1}$. Then,

$$gahg^{-1} = gag^{-1}ghg^{-1} = gag^{-1}h',$$

for some $h' \in H$ because H is normal in G . Therefore the map is well-defined since we have $g * aH = g(aH)g^{-1} = (gag^{-1})H \in G/H$. Notice that

$$1 * aH = (1a1^{-1})H = aH.$$

Moreover, for $g_1, g_2 \in G$, we have

$$\begin{aligned} g_1 * (g_2 * aH) &= g_1 * (g_2 a g_2^{-1})H = (g_1 g_2 a g_2^{-1} g_1^{-1})H = (g_1 g_2 a (g_1 g_2)^{-1})H \\ &= (g_1 g_2) * aH. \end{aligned}$$

Thus conjugation by G is a group action on G/H .

3. Prove that $*$: $\text{GL}_n(\mathbb{R}) \times \mathbb{R} \rightarrow \mathbb{R}$ defined by $A * x = \det(A)x$ is a group action.

First, the group action is well-defined since $\det(A)x$ is a product of real numbers and is thus a real number itself. The identity in $\text{GL}_n(\mathbb{R})$ is the $n \times n$ identity matrix I . We know that $\det(I) = 1$, thus $I * x = \det(I)x = 1x = x$, for any $x \in \mathbb{R}$. Now let $A, B \in \text{GL}_n(\mathbb{R})$, we have

$$A * (B * x) = A * (\det(B)x) = \det(A) \det(B)x = \det(AB)x = (AB) * x.$$

Therefore, $*$ is a group action.

4. Construct a group action $*$: $G \times \text{Aut}(G) \rightarrow \text{Aut}(G)$, where $\text{Aut}(G)$ is the group of automorphisms of G .

Consider the action $g * \varphi = \sigma_g \varphi$, where σ_g represents the inner automorphism $\sigma_g(x) = gxg^{-1}$. Since $\sigma_g \varphi$ is a composition of two automorphisms, it is also an automorphism and so $*$ is well-defined. Note that $\sigma_1 \varphi(x) = 1\varphi(x)1^{-1} = \varphi(x)$ and so $1 * \varphi = \sigma_1 \varphi = \varphi$. Take $g_1, g_2 \in G$, then

$$\sigma_{g_1} \sigma_{g_2} \varphi(x) = g_1 g_2 \varphi(x) g_2^{-1} g_1^{-1} = g_1 g_2 \varphi(x) (g_1 g_2)^{-1} = \sigma_{g_1 g_2} \varphi(x).$$

Thus $g_1 * (g_2 * \varphi) = \sigma_{g_1} \sigma_{g_2} \varphi = \sigma_{g_1 g_2} \varphi = g_1 g_2 * \varphi$. Therefore, $*$ is a group action by G on $\text{Aut}(G)$.

Or you could be boring and use the trivial group action $\odot$.

Something to think about: Would $g * \varphi = \varphi \sigma_g$ also be a group action?

5. Let G be a group acting on a set X .

- (a) Prove that, for all $x \in X$, the stabiliser $S(x)$ is a subgroup of G .
- (b) Prove that if, for $x, y \in X$, there is some $g \in G$ such that $y = g * x$, then $S(y) = gS(x)g^{-1}$.

(a) First note that because $1 * x = x$, we have $1 \in S(x)$, so $S(x)$ is nonempty. Take $a, b \in S(x)$. We then have

$$(ab) * x = a * (b * x) = a * x = x.$$

Therefore, $ab \in S(x)$. Moreover, we have

$$x = 1 * x = (a^{-1}a) * x = a^{-1} * (a * x) = a^{-1} * x.$$

Thus $a^{-1} \in S(x)$. Therefore, $S(x)$ is a subgroup of G .

(b) We need will first show that $S(y) \subseteq gS(x)g^{-1}$. Note that the question fixes g , it is not an arbitrary element of G . Take $g_y \in S(y)$. Then $g_y * y = y$. Using the fact that $y = g * x$, we then have $g_y g * x = g * x$. By Question 1 (b), this means that $g^{-1} g_y g * x = x$. That is, $g^{-1} g_y g \in S(x)$ and so $g_y = g(g^{-1} g_y g)g^{-1} \in gS(x)g^{-1}$. Thus $S(y) \subseteq gS(x)g^{-1}$.

Now we will show that $gS(x)g^{-1} \subseteq S(y)$. That $gg_xg^{-1} \in gS(x)g^{-1}$, where $g_x \in S(x)$. By Question 1 (b), we have $1 * y = g * x$ and so $g^{-1} * y = x$. Thus,

$$gg_xg^{-1} * y = gg_x * x = g * x = y.$$

Thus $gg_xg^{-1} \in S(y)$ and so $gS(x)g^{-1} \subseteq S(y)$. This allows us to conclude that $S(y) = gS(x)g^{-1}$, as desired.

6. Consider the additive group $\mathbb{R}$.

- (a) Prove that the map $*$: $\mathbb{R} \times \mathbb{C} \rightarrow \mathbb{C}$ defined by $a * z = ze^{ia}$, for $a \in \mathbb{R}$ and $z \in \mathbb{C}$, is a group action.
- (b) Find all stabilisers and orbits.
- (c) Geometrically, what does this action represent?

(a) The map is well-defined since $a * z = ze^{ia}$ is a product of two complex numbers, which is again a complex number. Note that since $\mathbb{R}$ is an additive group, the identity is 0. Then we have $0 * z = ze^0 = z$. For $a, b \in \mathbb{R}$, we have

$$a * (b * z) = a * ze^{ib} = ze^{ib}e^{ia} = ze^{i(a+b)} = (a + b) * z.$$

Thus $*$ is a group action.

(b) Let $z \in \mathbb{C}$. Note that if $z = 0$, then $a * z = 0 = z$ for all $a \in \mathbb{R}$. Thus $S(0) = \mathbb{R}$. On the other hand, if $z \neq 0$, then $a \in S(z)$ if and only if $a = 2\pi n$ for some $n \in \mathbb{Z}$. Therefore,

$$S(z) = \{2\pi n : n \in \mathbb{Z}\}.$$

Now, $w \in \mathbb{R} * z$ if and only if $w = ze^{ia}$ for some $a \in \mathbb{R}$. That is,

$$\mathbb{R} * z = \{ze^{ia} : a \in \mathbb{R}\}.$$

(c) The group action represents rotating a point in the complex plane by a radians around the origin. The orbit of a point $z \in \mathbb{C}$ is a circle with radius $|z|$.

Extra problems

For a nonempty set X , the set S_X , that consists of all bijections $\sigma : X \rightarrow X$, is a group under composition.

- (I) Let G be a group acting on a set X . For each $g \in G$, we can define a map $\sigma_g : X \rightarrow X$ by $\sigma_g(x) = g * x$. Prove that σ_g is a bijection. That is, $\sigma_g \in S_X$.

Take $g \in G$. Suppose $\sigma_g(x) = \sigma_g(x')$ for some $x, x' \in X$. That is, $g * x = g * x'$. Since $g \in G$, there exists the inverse $g^{-1} \in G$. We then have

$$x = 1 * x = (g^{-1}g) * x = g^{-1} * (g * x) = g^{-1} * (g * x') = (g^{-1}g) * x' = 1 * x' = x'.$$

Thus, σ_g is injective. Now take $x \in X$. We have that $x = 1 * x = (gg^{-1}) * x = g * (g^{-1} * x)$. By definition of the group action, $g^{-1} * x \in X$, that is $g^{-1} * x = x'$ for some $x' \in X$. Therefore, $x = g * x' = \sigma_g(x')$. We can now conclude that σ_g is a bijection on X .

- (II) Let G be a group acting on a set X . Prove that the map $\rho : G \rightarrow S_X$ defined by $\rho(g) = \sigma_g$ is a homomorphism.

By (I), ρ is well-defined. That is, $\rho(g) = \sigma_g$ is indeed an element of S_X for all $g \in G$. Take $g, g' \in G$. Then $\rho(gg') = \sigma_{gg'}$ and $\rho(g)\rho(g') = \sigma_g\sigma_{g'}$. We will show that these maps are the same by showing $\sigma_{gg'}(x) = \sigma_g\sigma_{g'}(x)$ for all $x \in X$. Taking $x \in X$, we have

$$\sigma_{gg'}(x) = gg' * x = g * (g' * x) = g * \sigma_{g'}(x) = \sigma_g(\sigma_{g'}(x)) = \sigma_g\sigma_{g'}(x).$$

Thus, ρ is a homomorphism.

- (III) Prove the following theorem:

Cayley's Theorem. *Every group G is isomorphic to a subgroup of S_G .*

HINT: Construct a group action and then use (II).

We have previously shown that the map $*$: $G \times G \rightarrow G$ defined by $g * x = gx$ for all $g, x \in G$ is a group action by G on G . By (II), we have a homomorphism $\rho : G \rightarrow S_G$. If we show that this is a monomorphism, then $G \cong G/\ker \rho \cong \text{im } \rho$, which is a subgroup of S_G . Suppose that $\rho(g) = \rho(g')$ for some $g, g' \in G$. That is, $\sigma_g(x) = \sigma_{g'}(x)$ for all $x \in G$, which means that $gx = g'x$. Since x is in the group G , we can multiply both sides by x^{-1} on the right to get $g = g'$. Thus, ρ is a monomorphism and so G is isomorphic to a subgroup of S_G .